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Investigating Clarinet Articulation Using a Physical Model and an Artificial Blowing Machine

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Summary
A time-domain physical model is presented that is capable of simulating a variety of articulation techniques in single-reed woodwind instruments and suitable for real-time sound synthesis. Due to the nonlinear nature of the excitation mechanism, an energy-based approach is adopted for the construction of the numerical scheme in order to ensure algorithm stability. To validate the model, measurements are carried out using an artificial blowing machine. The construction of the machine, including a sensor-equipped reed and mouthpiece as well as an automated artificial tongue and lip, is described in detail. By adjusting the motion of the tongue, the blowing machine can generate audio signals corresponding to portato and staccato articulation. These signals are resynthesised following an inverse modelling approach based on the presented physical model, during which model parameters are estimated. All estimated parameters lie in a physically feasible range and may be used for sound synthesis and sound analysis applications.

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1. Introduction

The physics of woodwind instruments has been extensively studied since the past century [1, 2, 3] and a deep understanding of their function has been obtained through both theoretical formulations and computer-aided simulations. Most of the processes that take place during the generation of musical tones are now well understood and may be reproduced using physical modelling techniques [4, 5, 6, 7, 8, 9, 10, 11, 12, 13].

Of particular interest is the control of the player over the instrument and how certain actions can be used to manipulate the resulting oscillations. In the case of single-reed woodwind instruments, a major part of this player-instrument interaction takes place at the mouthpiece. There, the embouchure of the player, along with the reed-mouthpiece system, constitutes a non-linear excitation mechanism that is coupled with a resonance tube, the function of which can be approximated using linear acoustics [7]. During note onsets, offsets and expressive performance, musicians use various articulation techniques by changing their embouchure, in order to control the oscillations of the instrument [14]. In effect, transient phenomena arise either at the beginning or the end of a note, or during tone transitions. Of particular interest in this study are those phenomena that are caused by the action of the player's tongue.

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The short duration of the transients and the fact that changes in the underlying physical quantities are hard to measure under real playing conditions, have confined most researchers to the study of steady-state tones. However, since the emerging transients are very important when it comes to identifying or evaluating musical instruments [15, 16, 17, 18, 19], they need to be accurately generated during physically-based sound synthesis. This requires both the formulation of time-domain physical models that are able to capture such transient effects as well as experimental measurements that will provide the necessary information to validate and update the above models.

Despite the fact that articulation is of primary importance for the musical quality of the generated sounds, most state-of-the-art modelling approaches use rather simple excitation mechanisms [20], due to the lack of relevant information on the tongue-reed interaction. Recent studies have tried to shed some light in this area using experimental measurements [21, 22] and physical modelling [11, 23] but a complete physics-based analysis is yet to be developed.

This work aims to formulate a model that captures all relevant physical phenomena that take place during tongue articulation. An energy balance approach is adopted in order to account for all energy source and loss mechanisms, while ensuring the numerical stability of the model [24, 25, 26, 27]. The model is then validated using a series of experiments carried out on an artificial blowing machine. Both the modelling and the experiments with the ar-

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tificial player allow the direct and continuous observation of the physically meaningful parameters that control the sound excitation mechanism; in terms of physical modelling these are directly available to the user, whereas in the case of the blowing machine they are continuously monitored by sensors. Including an automated artificial tongue in the blowing machine and explicitly modelling its interaction with the reed in the physical model renders this work suitable for studying various articulation techniques on single-reed woodwind instruments, such as staccato (an individual note separated from its neighbouring notes by silence) and portato (the notes are generally sustained, using tonguing to achieve some separation) [28, 29].

Section 2 presents the model of the player-instrument interaction, with special attention on numerical stability as an outcome of the energy-based approach that is followed. Discretisation is performed in Section 3 using finite difference time-domain methods. The energy balance of the coupled system is demonstrated both in the continuous and in the discrete domain. Section 4 presents the artificial blowing machine that has been used in order to obtain the experimental data for this study. A comparison between numerically synthesised and experimentally obtained sounds is carried out in Section 5. Nonlinear optimisation is employed in order to estimate the physical model parameters from the signals recorded using the artificial blowing machine. Finally Section 6 discusses the outcomes of this study in the context of music acoustics research.

2. Physical Modelling

Time-domain models of woodwind instruments typically consist of a nonlinear excitation mechanism coupled to a linear resonator [30]. In an attempt to better approximate wave propagation inside an axisymmetric tube without toneholes, Bilbao *et al.* [31] proposed a time-domain formulation of viscothermal losses making use of rational function approximations, as previously presented in [32]. A modified version of this formulation, using a semi-analytic technique for incorporating time-domain losses, is included in the present study. A time-domain representation is necessary here, since the aim is to simulate transient phenomena. Regarding the excitation mechanism, the mouthpiece-reed system is represented by a single mass-spring model including (i) a collision term that models the beating of the reed on the mouthpiece lay and (ii) a collision term that models the tongue-reed interaction.

Both the excitation mechanism and the resonator are numerically treated using the finite difference method [24]. In order to ensure the stability of the developed algorithms, an energy balance is obtained for the continuous models in terms of the rate of change of the energy stored in the system (the acoustic tube and the vibrating reed). In the discrete domain this energy balance can be used to bound the approximate solutions, thus ensuring numerical stability of the developed schemes [33, 31].

2.1. Viscothermal Wave Propagation

Consider an axisymmetric tube of length L and variable cross-sectional area $S = S(x), 0 \leq x \leq L$. The propagation of plane waves in tubular ducts, taking viscothermal losses into account, can be modelled in the frequency domain by [31, 34]

$$\partial_x P + ZV = 0, \tag{1a}$$
$$\partial_x (SV) + YSP = 0, \tag{1b}$$

where $P = P(\omega, x)$ is the acoustic pressure and $V = V(\omega, x)$ the particle velocity, ω denoting the angular frequency. The functions Z and Y represent the specific series impedance and the shunt admittance respectively and are given by

$$Z(\omega, x) = \frac{\mathrm{i}\omega\rho}{1 - F_v}, \quad Y(\omega, x) = \frac{\mathrm{i}\omega}{\rho c^2}(1 + (\gamma - 1)F_\theta), \tag{2}$$

where i denotes the imaginary unit, c is the speed of sound, ρ is air density and γ is the heat capacity ratio. F_v and F_θ are defined as

$$F_v = \phi(\sqrt{-\mathrm{i}} r_v), \quad F_\theta = \phi(\sqrt{-\mathrm{i}} r_\theta), \quad \phi(\zeta) = \frac{2J_1(\zeta)}{\zeta J_0(\zeta)}. \tag{3}$$

J_0 and J_1 denote the Bessel functions of order zero and one respectively and the viscous and thermal boundary layer thicknesses r_v and r_θ are given by [35, 36]

$$r_v = \sqrt{\frac{S\rho\omega}{\pi\eta}}, \qquad r_\theta = \nu r_v, \tag{4}$$

where $\eta = 1.82 \times 10^{-5}$ kg/(m·s) is the viscosity coefficient and $\nu = 0.842$ is the square root of the Prandtl number.

Following the approach proposed in [31], model (1) is transformed to the time domain by first splitting Z and Y as

$$Z = \mathrm{i}\omega\rho + Z_v, \tag{5a}$$
$$Y = \frac{\mathrm{i}\omega}{\rho c^2} + Y_\theta. \tag{5b}$$

Since the cross-sectional area S does not depend on the time t, (1) transforms into

$$\partial_x p + \rho \partial_t v + z_v * v = 0, \tag{6a}$$
$$\partial_x (Sv) + \frac{S}{\rho c^2}\partial_t p + S y_\theta * p = 0, \tag{6b}$$

where $*$ denotes convolution with respect to the time variable t, and lower case function names refer to the time-domain versions of the capitalised functions. The boundary conditions in the time domain are given by $v(t, 0) = v_{\mathrm{in}}(t)$ at $x = 0$, where v_{in} is particle velocity inside the mouthpiece, and

$$p(t, L) = S(L) z_r * v(t, L) \tag{7}$$

at $x = L$, z_r being the time-domain counterpart of a given radiation impedance Z_r.

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Reference [37] proposes to calculate the time-domain convolutions $z_v * v$ and $y_\theta * p$ by approximating Z_v and Y_θ in the frequency domain by rational functions Z_v^K and Y_θ^K, as

$$Z_v^K = R_0 + \sum_{k=1}^{K} \frac{R_k \mathrm{i}\omega}{D_k + \mathrm{i}\omega}, \quad Y_\theta^K = \sum_{k=1}^{K} \frac{G_k \mathrm{i}\omega}{C_k + \mathrm{i}\omega}. \tag{8}$$

One can show that $Y_\theta(0, x) = 0$, so a constant term in the approximation of Y_θ is not needed. R_k, D_k, G_k and C_k (all non-negative due to passivity requirements [31]) can be found by minimizing the differences $Z_v - Z_v^K$ and $Y_\theta - Y_\theta^K$ on a fixed set of frequencies (in this case the hearing range) by Newton's method [31]. Note that the coefficients depend on the radius at the spatial variable x, but for simplicity the x-dependency is here omitted. This yields

$$z_v * v \approx R_0 v + \sum_{k=1}^{K} w_k \quad \text{and} \quad y_\theta * p \approx \sum_{k=1}^{K} q_k, \tag{9}$$

where the frequency-domain versions of w_k and q_k are given by the relations

$$W_k = \frac{R_k \mathrm{i}\omega}{D_k + \mathrm{i}\omega} V \quad \text{and} \quad Q_k = \frac{G_k \mathrm{i}\omega}{C_k + \mathrm{i}\omega} P. \tag{10}$$

A semi-analytical formulation was recently proposed by the authors for the calculation of w_k and q_k [38]. For $\xi > 0$ and $\vartheta(t)$ denoting the Heaviside step function, the Fourier transform of $\vartheta(t)\mathrm{e}^{-\xi t}$ is $\dfrac{1}{\xi + \mathrm{i}\omega}$. It follows then from (10) that

$$w_k(t) = R_k \int_0^t \mathrm{e}^{-D_k(t-\tau)} \partial_t v(\tau)\, \mathrm{d}\tau, \tag{11a}$$

$$q_k(t) = G_k \int_0^t \mathrm{e}^{-C_k(t-\tau)} \partial_t p(\tau)\, \mathrm{d}\tau. \tag{11b}$$

Hence, using the above substitution, the wave propagation inside the tube can be modelled as

$$\partial_x p + \rho \partial_t v + R_0 v + \sum_{k=1}^{K} w_k = 0, \tag{12a}$$

$$\partial_x (S v) + \frac{S}{\rho c^2} \partial_t p + S \sum_{k=1}^{K} q_k = 0. \tag{12b}$$

In a numerical method, the number of rational functions that can be used is limited. We observed experimentally, that for $D > 0$, only a finite number of functions of the form

$$F_D = \frac{\mathrm{i}\omega}{D + \mathrm{i}\omega} \tag{13}$$

on a given interval are numerically linearly independent. This means that adding further functions to the basis can not increase precision. However in the present case, considering only audible frequencies, it has been observed in [31] that 16 functions are approximating the impedance Z_v with a fractional error of 10^{-4}.

Boundary conditions need to be specified for both ends of the tube. At the mouthpiece end ($x = 0$) the tube is coupled to the player's embouchure (see Section 2.2). At the radiating end ($x = L$) the radiation impedance Z_r can be approximated by [39, 24]

$$Z_r(\omega) = \frac{\mathrm{i}\omega R_r}{D_r + \mathrm{i}\omega}. \tag{14}$$

The coefficients R_r and D_r depend on the radius and moreover on the flange of the tube [24]. Further, more refined versions are discussed in [40]. With this choice of approximation, the boundary condition at $x = L$ in the time domain is, because of (7), given by the differential equation

$$S(L) R_r \partial_t v(t, L) = D_r p(t, L) + \partial_t p(t, L). \tag{15}$$

2.2. Excitation Mechanism

On the driving-end of a woodwind instrument a vibrating reed acts as a pressure controlled valve driven by the difference between the blowing pressure p_b and the internal mouthpiece pressure $p_\mathrm{in} = p(t, 0)$, namely $p_\Delta = p_\mathrm{b} - p_\mathrm{in}$. The reed can be modelled as a single mass-spring system [41] where the force applied due to the pressure difference p_Δ is $f_\Delta = S_\mathrm{r} p_\Delta$, S_r being the effective reed area [41, 13]. Two nonlinear interactions take place here: one due to the collision of the reed with the mouthpiece lay and one due to the tongue-reed interaction. These can be modelled using the Hunt-Crossley impact model [42], which results in a force of the form [43]

$$f_\mathrm{c} = -\frac{\partial V_\mathrm{c}}{\partial y} - r \frac{\partial V_\mathrm{c}}{\partial t}, \tag{16}$$

where r is a damping coefficient and

$$V_\mathrm{c} = \frac{k_\mathrm{c}}{\alpha + 1} \lfloor y \rfloor^{\alpha+1} \tag{17}$$

is the collision potential, k_c being the collision stiffness [26]. $\lfloor y \rfloor^\alpha = \vartheta(y) y^\alpha$, where $\vartheta(y)$ denotes the Heaviside step function and $\alpha \geq 1$ is a power-law exponent. This non-negative term represents the compression of the colliding objects while they are in contact [26]. Also considering that the player's tongue is moving, the motion of the vibrating reed can be described, in a lumped-model approximation [11, 41, 12, 13] by

$$\begin{aligned} m \frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + m\gamma \frac{\mathrm{d}y}{\mathrm{d}t} + k y \\ - k_\mathrm{tg} \lfloor y_\mathrm{tg} - y \rfloor^{\alpha_\mathrm{tg}} \left(1 + r_\mathrm{tg} \frac{\mathrm{d}(y_\mathrm{tg} - y)}{\mathrm{d}t} \right) \\ + k_\mathrm{lay} \lfloor y - y_\mathrm{lay} \rfloor^{\alpha_\mathrm{lay}} \left(1 + r_\mathrm{lay} \frac{\mathrm{d}y}{\mathrm{d}t} \right) = S_\mathrm{r} p_\Delta, \end{aligned} \tag{18}$$

where m, k and γ are the effective mass, effective stiffness, and effective damping of the reed respectively, considering the action of the player's lip. y is the displacement from the reed resting position ($y = 0$) after the player positions his/her lip [11]. The parameters related to contact

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modelling (contact stiffness k_κ, contact position y_κ, collision exponent α_κ, contact damping r_κ and later also collision potential $V_{c,\kappa}$) correspond to reed-mouthpiece collision ($\kappa = \text{lay}$) and tongue-reed interaction ($\kappa = \text{tg}$). Defining the kinetic energy of the reed $T = \varphi^2/(2m)$, φ being the conjugate momentum, and the stiffness potential $V_s = ky^2/2$ leads to a Hamiltonian formulation of the system, namely

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{\partial T}{\partial \varphi}, \tag{19a}$$

$$\frac{\mathrm{d}\varphi}{\mathrm{d}t} = -\frac{\partial V}{\partial y} - r_{\text{lay}}\frac{\partial V_{\text{c,lay}}}{\partial t} - \gamma\varphi + f_{\text{ex}}(y, \varphi, t), \tag{19b}$$

where $V = V_s + V_{\text{c,lay}}$. Note that the effect of the tongue-reed interaction is included in the external force, which is defined as $f_{\text{ex}}(y, \varphi, t) = f_{\text{tg}}(y, \varphi, t) + f_\Delta(y, \varphi, t)$, with

$$f_{\text{tg}}(y, \varphi, t) = k_{\text{tg}}\lfloor y_{\text{tg}} - y\rfloor^{\alpha_{\text{tg}}}\left(1 + r_{\text{tg}}\left(\frac{\mathrm{d}y_{\text{tg}}}{\mathrm{d}t} - \frac{\varphi}{m}\right)\right).$$

A further nonlinearity takes place here due to the relationship between the mouthpiece pressure p_{in} and the mouthpiece flow u_{in}. The flow is built up from two components [7], the Bernoulli flow u_{f} and the flow u_{r} induced by the motion of the reed, with

$$u_{\text{in}} = u_{\text{f}} + u_{\text{r}}, \tag{20a}$$

$$u_{\text{f}} = \sigma\lambda h\sqrt{\frac{2|p_\Delta|}{\rho}}, \tag{20b}$$

$$u_{\text{r}} = S_{\text{r}}\frac{\mathrm{d}y}{\mathrm{d}t}, \tag{20c}$$

where $\sigma = \text{sign}(p_\Delta)$, λ is the width of the reed and $h = \lfloor y_{\text{eq}} - y\rfloor$ is the reed opening, y_{eq} being the reed opening at rest (equilibrium opening). These parameters, related to the single-reed excitation mechanism, are visualised in Figure 1.

2.3. Energy Balance

In terms of the total energy, there are two different ways to analyse the system. One is to consider the player as an external source of energy that is driving the (passive) instrument. Another useful approach is to split linear from nonlinear parts, thus observing the system from the viewpoint of the reed. Upstream lies the nonlinear excitation mechanism (including Bernoulli flow, reed beating and tongue-reed interaction) and downstream lies the linear resonator (including viscothermal and radiation losses).

Regarding the viscothermal wave propagation model, following [31], (12a) is multiplied by Sv and (12b) by p to obtain the energy variation in the tube

$$\frac{\mathrm{d}\mathcal{H}_{\text{T}}}{\mathrm{d}t} = u(0)p(0) - Q_v - Q_\theta - p(L)u(L), \tag{21}$$

where

$$Q_v = R_0\int_0^L Sv^2\,\mathrm{d}x + \sum_{k=1}^{K}\int_0^L S\frac{w_k^2}{R_k}\,\mathrm{d}x, \tag{22a}$$

$$Q_\theta = \sum_{k=1}^{K}\int_0^L S\frac{q_k^2}{G_k}\,\mathrm{d}x, \tag{22b}$$

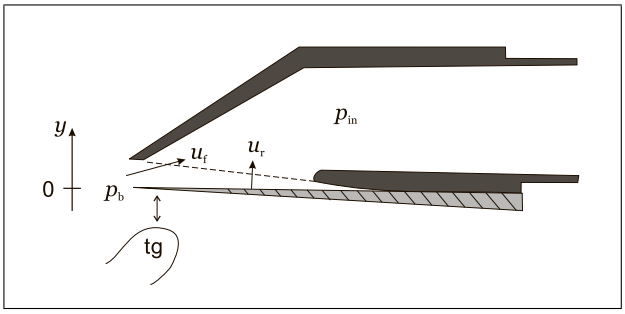


Figure 1. Sketch of the single-reed–mouthpiece system; "tg" represents the player's tongue, u_{f} and u_{r} the Bernoulli and reed flow respectively and p_{b} and p_{in} the blowing pressure and mouthpiece pressure.

and $p(0) = p_{\text{in}}$, $u(0) = u_{\text{in}}$ are the pressure and the volume flow inside the mouthpiece.

Regarding the excitation mechanism, for the evolution of the total energy of the reed $\mathcal{H}_{\text{R}} = T + V$, the following expression can be derived,

$$\frac{\mathrm{d}\mathcal{H}_{\text{R}}}{\mathrm{d}t} = \frac{\partial V}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{\partial T}{\partial \varphi}\frac{\mathrm{d}\varphi}{\mathrm{d}t} = Q_{\text{r}} + Q_{\text{ex}}, \tag{23}$$

with

$$Q_{\text{r}} = -\frac{\varphi^2}{m^2}\left(m\gamma + r_{\text{lay}}k_{\text{lay}}\lfloor y - y_{\text{lay}}\rfloor^{\alpha_{\text{lay}}}\right)$$

and

$$Q_{\text{ex}} = \frac{\varphi}{m}\left(f_{\text{tg}}(y, \varphi, t) + f_\Delta(y, \varphi, t)\right).$$

Q_{r} signifies the rate at which energy is dissipated due to both types of friction and is always positive, while Q_{ex} signifies energy variation due to the applied forces and may be of arbitrary sign. A similar energy balance is derived in [33] directly from Newton's equation of motion, and is here extended to include impact damping and tongue-reed interaction. Multiplying (18) by $\mathrm{d}y/\mathrm{d}t$, and using (20), one may arrive at the following energy balance for the reed:

$$\frac{\mathrm{d}\mathcal{H}_{\text{R}}}{\mathrm{d}t} = -Q_{\text{r}} - Q_{\text{in}} - p_{\text{in}}u_{\text{in}} + p_{\text{b}}u_{\text{in}} + \frac{\varphi}{m}f_{\text{tg}}(y, \varphi, t), \tag{24}$$

where $Q_{\text{in}} = \lambda h(2/\rho)^{-1/2}|p_\Delta|^{3/2} \geq 0$ represents the energy lost due to the pressure drop in the mouthpiece. This is identical to (23) as shown in Appendix A1. Furthermore, mass continuity dictates that the Bernoulli flow is identical across the reed (i.e. in the player's mouth and in the mouthpiece). The same is true for the reed induced flow [44]; a closing reed generates a flow towards the bore on both sides of the reed (inside the player's mouth and inside the mouthpiece), while an opening reed generates a flow "away" from the mouthpiece (these flows being of equal magnitude, at least in this lumped approximation). Hence, it can be deduced that $u_{\text{in}} = u_{\text{b}}$, where u_{b} is the total flow inside the player's mouth.

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Combining equations (21) and (24) and defining the radiation losses $Q_L = p_L u_L$, the evolution of the total system energy $\mathcal{H} = \mathcal{H}_\mathrm{T} + \mathcal{H}_\mathrm{R}$ is given by

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} = p_\mathrm{b} u_\mathrm{b} - Q_v - Q_\theta - Q_\mathrm{r} - Q_\mathrm{in} - Q_L + \frac{\varphi}{m} f_\mathrm{tg}, \qquad (25)$$

which, as shown in Appendix A1 is identical to

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} = -Q_v - Q_\theta - Q_\mathrm{r} - Q_L + \frac{\varphi}{m} f_\mathrm{ex} + p_\mathrm{in} u_\mathrm{in}. \qquad (26)$$

As discussed above, equations (25) and (26) offer two alternative representations of the energy evolution. The former is equal to mechanical losses in the reed and viscothermal plus radiation losses ($-Q_\mathrm{r} - Q_v - Q_\theta - Q_L$) plus loss due to pressure drop in the mouthpiece ($-Q_\mathrm{in}$) plus power input from the player's blowing pressure ($p_\mathrm{b} u_\mathrm{b}$) and tongue motion ($f_\mathrm{tg}\,\varphi/m$). Hence, a passive system is driven by the external forces provided by the player. The latter formulation (26), which has also been used to study numerical stability, is seen from the point of view of the reed. The energy balance corresponds to viscothermal, radiation and mechanical losses ($-Q_v - Q_\theta - Q_L - Q_\mathrm{r}$) plus power input from the excitation mechanism to the tube ($p_\mathrm{in} u_\mathrm{in}$) plus energy variation due to the external force applied on the reed ($f_\mathrm{ex}\,\varphi/m$). This force here stems from both the tongue action and the pressure difference across the reed. Thus the instrument is here split into a nonlinear element (embouchure plus vibrating reed) and a linear element (acoustic tube).

Note that the last two terms on the right hand side of (26) are of arbitrary sign. In the absence of boundary excitation ($p_\mathrm{b} = p_L = 0$) and zero tongue force ($f_\mathrm{tg} = 0$) their sum is equal to $-u_\mathrm{f} p_\Delta$ (see Equation (A3)), which is always non-positive (from (20b)); hence, the system is dissipative, as also expected from (25). Despite energy dissipation, it is still possible to arrive at an energy-conservation law, by adding up the energy that is lost at each time step, as proposed in [45] and subsequently adopted in passive time-domain models of musical instruments [46, 31]. Thus, in the presence of excitation, a conservation law can be derived in the form of

$$K = \mathcal{H} + \int \left(Q_v + Q_\theta + Q_\mathrm{r} + Q_L - \frac{\varphi}{m} f_\mathrm{ex} - p_\mathrm{in} u_\mathrm{in} \right) \mathrm{d}t = \text{const.} \qquad (27)$$

that can be used to ensure stability of the proposed numerical scheme [24, 33].

3. Discretisation

In order to obtain an approximate solution for the wave propagation model, Equation (12) needs to be discretised. Approximations p_l^n and v_l^n to the solutions p and v of (12) respectively, are computed at discrete points (t_n, x_l) in time and space, where $t_n = n\Delta_t$, $n = 0, 1, 2 \ldots$ and $x_l = l\Delta_x$ for $l = 0, \ldots, M$ and $L = M\Delta_x$, for fixed Δ_t and Δ_x. One iteratively computes p_l^{n+1} and v_l^{n+1} from results obtained at previous time steps. In order to derive a discrete model, a discretisation of (11) is also required, in the form of

$$w_{k,l}^{n+1} = \mathrm{e}^{-D_{k,l}\Delta_t} w_{k,l}^n + R_{k,l}\left(v_l^{n+1} - v_l^n\right)\mathrm{e}^{-D_{k,l}\frac{\Delta_t}{2}}, \qquad (28a)$$

$$q_{k,l}^{n+1} = \mathrm{e}^{-C_{k,l}\Delta_t} q_{k,l}^n + G_{k,l}\left(p_l^{n+1} - p_l^n\right)\mathrm{e}^{-C_{k,l}\frac{\Delta_t}{2}}. \qquad (28b)$$

Equation (12a) is discretised by finite differences and, using (28a), one can compute $v_l^{n+1}, l = 1, \ldots, M$ from

$$\frac{p_l^n - p_{l-1}^n}{\Delta_x} + \rho\frac{v_l^{n+1} - v_l^n}{\Delta_t} + R_{0,l} v_l^{n+1} + \sum_{k=1}^{K}\left[\mathrm{e}^{-L_{k,l}\Delta_t} w_{k,l}^n + R_{k,l}(v_l^{n+1} - v_l^n)\mathrm{e}^{-L_{k,l}\frac{\Delta_t}{2}}\right] = 0. \qquad (29)$$

Discretising (12b), and using (28b), one computes p_l^{n+1}, $l = 1, \ldots, M-1$ from

$$\frac{S_{l+1} v_{l+1}^{n+1} - S_l v_l^{n+1}}{\Delta_x} + \frac{S_l}{\rho c^2}\frac{p_l^{n+1} - p_l^n}{\Delta_t} + S_l \sum_{k=1}^{K}\left[\mathrm{e}^{-C_{k,l}\Delta_t} q_{k,l}^n + G_{k,l}(p_l^{n+1} - p_l^n)\mathrm{e}^{-C_{k,l}\frac{\Delta_t}{2}}\right] = 0. \qquad (30)$$

Finally $w_{k,l}^{n+1}$ and $q_{k,l}^{n+1}$ are updated by (28a) and (28b), respectively. For the right-hand boundary ($l = M$) taking finite differences in (15) yields

$$S_M R_r \frac{v_M^{n+1} - v_M^n}{\Delta_t} = D_r p_M^{n+1} + \frac{p_M^{n+1} - p_M^n}{\Delta_t}, \qquad (31)$$

from which the pressure at the end of the tube p_M^{n+1} can be computed. For the left-hand boundary ($l = 0$) the discretisation of the excitation mechanism needs to be considered, as follows.

In order to establish a causal, computable loop, all variables at time $(n+1)$ need to be calculated by the known values of variables at time n and a set of initial conditions at $n = 0$. The discretised Hamilton's equations (19) take the form

$$\frac{y^{n+1} - y^n}{\Delta_t} = \frac{T(\varphi^{n+1}) - T(\varphi^n)}{\varphi^{n+1} - \varphi^n} = \frac{1}{2m}\left(\varphi^{n+1} + \varphi^n\right) \qquad (32a)$$

$$\frac{\varphi^{n+1} - \varphi^n}{\Delta_t} = -\frac{V(y^{n+1}) - V(y^n)}{y^{n+1} - y^n} - r_\mathrm{lay}\frac{V_\mathrm{c,lay}(y^{n+1}) - V_\mathrm{c,lay}(y^n)}{\Delta_t} - \gamma\frac{\varphi^{n+1} + \varphi^n}{2} + \frac{f_\mathrm{ex}^{n+1} + f_\mathrm{ex}^n}{2} \qquad (32b)$$

Defining

$$\psi^n = \varphi^n \frac{\Delta_t}{2m} \qquad (33)$$

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leads to solving a nonlinear equation in $s = y^{n+1} - y^n = \psi^{n+1} + \psi^n$, namely

$$\begin{aligned} F(s) = &\left(1 + \frac{\gamma\Delta_t}{2}\right) s - 2\psi^n \\ &+ \frac{\Delta_t^2}{2m}\left(\frac{V(y^n+s) - V(y^n)}{s}\right) \\ &+ \frac{r_{\text{lay}}\Delta_t}{2m}\left(V_{\text{c,lay}}(y^n+s)\right. \\ &\left. - V_{\text{c,lay}}(y^n)\right) - \frac{\Delta_t^2}{2m}\mu_{t+} f_{\text{ex}}^n = 0, \end{aligned} \tag{34}$$

where $\mu_{t+} f_{\text{ex}}^n = (f_{\text{ex}}^{n+1} + f_{\text{ex}}^n)/2$ and

$$\begin{aligned} f_{\text{ex}}^{n+1} &= S_{\text{r}} p_\Delta^{n+1}\left(y^{n+1}, q^{n+1}, t^{n+1}\right) + f_{\text{tg}}^{n+1}\left(y^{n+1}, q^{n+1}, t^{n+1}\right) \\ &= S_{\text{r}} p_\Delta\left(y^n + s, \tfrac{2m}{\Delta_t}(s - \psi^n), t^{n+1}\right) \\ &\quad + f_{\text{tg}}\left(y^n + s, \tfrac{2m}{\Delta_t}(s - \psi^n), t^{n+1}\right). \end{aligned} \tag{35}$$

After s is calculated (using an iterative method; see e.g. [43]) φ^{n+1}, y^{n+1}, and hence h^{n+1} can be updated. Regarding flow through the mouthpiece, discretising (20a) yields

$$S_0 v_0^{n+1} = 2\frac{S_{\text{r}}}{\Delta_t}\psi^{n+1} + \sigma\lambda h^{n+1}\sqrt{2/\rho}\sqrt{\left|p_\Delta^{n+1}\right|}. \tag{36}$$

Substituting (36) into (30) for $l = 0$ a nonlinear equation for p_0^{n+1} can be obtained that has the following form:

$$\begin{aligned} &- \sigma\sqrt{\left|p_\Delta^{n+1}\right|}\frac{\lambda h^{n+1}\sqrt{\frac{2}{\rho}}}{\Delta_x} \\ &- \sigma\left|p_\Delta^{n+1}\right|\left(\frac{S_0}{\Delta_t\rho c^2} + S_0\sum_{k=1}^{K}\left[G_{k,0}\mathrm{e}^{-C_{k,0}\frac{\Delta_t}{2}}\right]\right) \\ &+ (p_{\text{b}}^{n+1} - p_0^n) S_0 \sum_{k=1}^{K}\left[G_{k,0}\mathrm{e}^{-C_{k,0}\frac{\Delta_t}{2}}\right] \\ &+ S_0\sum_{k=1}^{K}\left[\mathrm{e}^{-C_{k,0}\Delta_t} q_{k,0}^n\right] \\ &+ \frac{S_1 v_1^{n+1} - 2\frac{S_{\text{r}}}{\Delta_t}\psi^{n+1}}{\Delta_x} + \frac{S_0}{\rho c^2}\frac{p_{\text{b}}^{n+1} - p_0^n}{\Delta_t} = 0. \end{aligned} \tag{37}$$

This may be written as

$$-K_1\sigma\sqrt{\left|p_\Delta^{n+1}\right|} - K_2\sigma\left|p_\Delta^{n+1}\right| + K_3 = 0. \tag{38}$$

Since the coefficients G_k are nonnegative, K_1 and K_2 are positive, hence

$$\sigma = \text{sign}(K_3). \tag{39}$$

This leads to a quadratic equation for $\sqrt{\left|p_\Delta^{n+1}\right|}$, with solutions

$$\frac{-K_1 \pm \sqrt{K_1^2 + 4K_2|K_3|}}{2K_2}. \tag{40}$$

The only positive solution is

$$\sqrt{\left|p_\Delta^{n+1}\right|} = \frac{-K_1 + \sqrt{K_1^2 + 4K_2|K_3|}}{2K_2} \tag{41}$$

and, since the sign of the pressure difference is already known, the value of v_0^{n+1} can be finally obtained from (36). A discrete conservation law follows in the form of

$$\begin{aligned} \mathcal{K}^n &= \text{const.} \\ &= \mathcal{H}_{\text{R}}^{n+1} - \sum_{j=0}^{n}\tfrac{\Delta_t}{m}\mu_{t+}\varphi^n\left(-r_{\text{lay}}\frac{V_{\text{c,lay}}^{j+1} - V_{\text{c,lay}}^j}{\Delta_t}\right. \\ &\quad \left. - \gamma\mu_{t+}\varphi^j + \mu_{t+} f_{\text{ex}}^j\right). \end{aligned} \tag{42}$$

Note that in the absence of an initial energy (which is the case here) this quantity remains equal to zero at all times. Errors in the conservation of $\mathcal{K}$ are hence quantified by the difference $\mathcal{K}^n - \mathcal{K}^0$ (here equal to $\mathcal{K}^n$). Pressure signals synthesised using the presented model are shown in Sec. 5.1, compared to signals obtained using the artificial blowing machine as described in the next section. The error in the conservation of $\mathcal{K}^n$ is also plotted, illustrating the numerical stability of the scheme.

4. Artificial Blowing Machine

Objective characterisation of sound excitation in musical instruments often requires the construction of artificial playing machines. Regarding single-reed woodwind instruments several such attempts have been recently reported (see [47] for a recent review). They may be roughly categorised in two classes—namely, devices operating at the driving end [10, 48, 49] (where air pressure is supplied through an artificial "oral cavity") and devices acting downstream of the reed [50, 51] (using either flow aspiration or loudspeaker excitation). The former constitute a better representation of human players, since the excitation of the instrument oscillations is achieved in the same manner as under real playing conditions, allowing a more realistic representation of transients.

4.1. Experimental Setup

The setup used in this paper (see Figure 2) follows the artificial blowing approach, where a 170 cm^3 transparent Plexiglas box acts as the oral cavity. Air-pressure is supplied from the bottom of the box through a proportional valve (PVQ33-5G-23, by SMC). A pressure sensor (Technoterm 5402) inside the box monitors the pressure inside the cavity (p_{b}) and allows for adjustments of the valve settings to be done either manually or automatically by a custom designed control program based on the LabView Software (Ver. 2011, by National Instruments) and Hardware Platform (PCIe-6351, by National Instruments). Through the valve and the control software, the system is capable of providing stable air pressure up to 10 kPa inside the box.

For measurements, a German B♭-clarinet is mounted through a circular opening at one side of the box in a way

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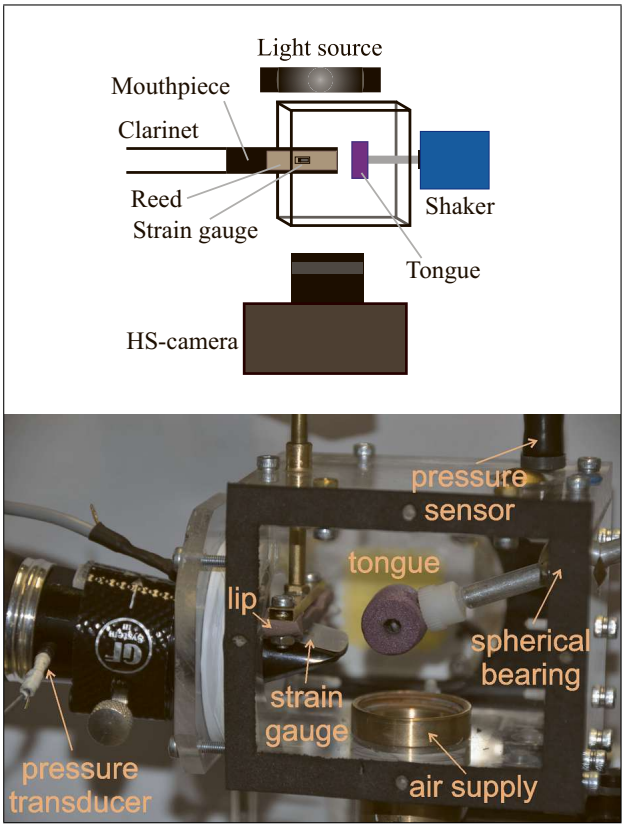


Figure 2. (Colour online) Top: Sketch of the experimental setup for the artificial blowing machine (view from above; not to scale), including a highspeed camera, a sensor-equipped reed and an artificial tongue mounted on a shaker. Bottom: Photo of the artificial mouth (view from the side).

that the reed (Légère, German cut, Strength 3.5) faces to the top and half of the mouthpiece (NA1 by Maxton) enters the box. The opening for the clarinet is sealed by a pneumatic air lock during operation. A $42 \times 5 \times 4\,\mathrm{mm}^3$ piece of synthetic polyester attached to a metal support is mounted on a translation stage (Thorlabs DTS50/M) and acts as an artificial lip to dampen the reed vibrations. A zoomed-in camera image is also used to verify the position of the lip. A strain gauge sensor on the clarinet reed captures the reed bending during sound production, while a pressure sensor (Endevco 8507C-2) inside the mouthpiece tracks the mouthpiece pressure (p_{in}).

An automated artificial tongue that consists of a cylinder of 1.3 cm diameter and 2.2 cm length made of cellular polyethylene is placed inside the box. It is attached to a rod mounted on a SmartShaker (Model TMS-K2007-E01, by PCB Group, Inc.) outside the box. The shaker can provide a force up to 31 N, a stroke length of up to 13 mm and supports DC control voltage, so that the tongue can remain in any position if required. The movement patterns of the shaker are adjusted within the LabView control program. Parameters for the duration of forward and backward motion (along the direction of the rod) as well as the duration for touching and releasing the reed are set (see Figure 4, top). The tongue motion is monitored using an acceleration sensor (288D01, by PCB Piezotronics, Inc.)

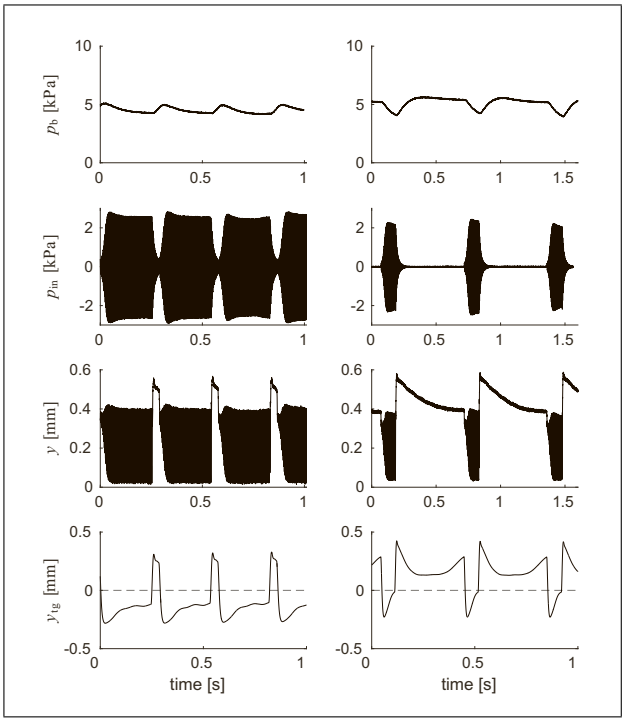


Figure 3. Blowing pressure (p_{b}), mouthpiece pressure (p_{in}), reed displacement (y) and tongue displacement (y_{tg}) obtained using the artificial blowing machine for portato (left) and staccato (right) articulation.

that quantifies the motion of the shaker, which drives the tongue-rod mechanism. These signals can be used to verify that the tongue was moving according to the instructions given (see bottom of Figures 3 and 4). During the experiment, the control program captures all sensor signals simultaneously at 50000 Hz (air supply pressure, mouthpiece pressure, reed bending, tongue acceleration).

Setting the shaker to move the tongue to the reed and remain in contact for a duration of 30 ms results in reed and mouthpiece pressure signals (see Figure 3, left) which are characteristic of portato articulation [23]. Modifying the duration of tongue-reed contact settings results in different tone lengths up to very short staccato tones (see Figure 3, right). Note that during tongue-reed contact the bending of the reed is severely affected by the action of the tongue, resulting in artificially large values, as also obtained in experiments carried out with real players [14, 23]. The transient signals obtained using the artificial blowing machine can be used as "target signals" for an inverse modelling approach to estimate physical model parameters, as described in the next section.

5. Sound Resynthesis

In contrast to previous inverse modelling attempts that were aiming at estimating parameters related only to the excitation mechanism [52, 53, 51] or only to the geometry of the bore [54, 55, 38], in this study both exciter and resonator are treated simultaneously. Furthermore, the estimation of the physical model parameters is not only car-

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ried out for steady-state conditions, but is hereby extended to note transitions. Such an analysis is now possible due to the experimental arrangement described in the previous section, which allows artificial tongue articulation while sensing the parameters related to the reed-mouthpiece system.

Two different fingerings are used for this experiment; one with none of the clarinet's toneholes closed that refers to a G4 fingering (concert pitch F4; 349 Hz) and one that refers to a D4 fingering (concert pitch C4; 261 Hz). Two different blowing pressures were provided (4500 Pa and 6500 Pa), hence the resulting measurements are labelled using two letters, one that refers to the note (G or D) and one that refers to low or high blowing pressure (p or P), i.e. Dp, DP, Gp and GP. The lip position remains unaltered throughout the experiments, whereas the lip force is manually adjusted to avoid overblowing.

Based on the captured signals the objective is to estimate reed parameters, the effective length of the resonator and the parameters related to the tonguing action. The Levenberg-Marquardt optimisation method is used for this purpose [56]. This is an iterative procedure that combines gradient descent and the Gauss-Newton algorithms. An objective function (to be minimised during the optimisation procedure) is constructed by considering the difference between the measured (target) signals and resynthesised signals (using the numerical model) for both the reed displacement and the mouthpiece pressure, namely

$$f_{\text{obj}} = \sqrt{\left(\frac{||\mathbf{y}-\hat{\mathbf{y}}||}{||\mathbf{y}||}\right)^2 + \left(\frac{||\mathbf{p}-\hat{\mathbf{p}}||}{||\mathbf{p}||}\right)^2}, \tag{43}$$

where $(\mathbf{y}, \mathbf{p})$ and $(\hat{\mathbf{y}}, \hat{\mathbf{p}})$ are the measured and resynthesised reed displacement and mouthpiece pressure signals respectively. Note that for the reed signal the part that corresponds to tongue-reed contact is omitted, since, as explained in the previous section, the bending signal is distorted by the action of the tongue. The measured blowing pressure is used to excite the model. In contrast to signals related to the lip and tongue of the player, the above signals may also be obtained by measurements carried out with real players in a non-intrusive manner [14].

The effective length of the resonator (L) is optimised first and the remaining model parameters are then added to the search space. The termination criterion for the optimisation algorithm is a relative change in the difference between two consecutive solutions smaller than 10^{-6}. The motion of the tongue during portato articulation is described as follows. The tongue position is taken to be constantly equal to -2×10^{-5} m outside the interval $[t_1, t_4]$ (corresponding to no tongue-reed contact), equal to $\max(y_{\text{tg}})$ inside the interval $[t_2, t_3]$ and linear on $[t_1, t_2]$ and $[t_3, t_4]$ (see Figure 4). For staccato articulation, since a single tone is synthesised (see below) the motion of the tongue is inverted.

5.1. Results

Starting values for the model parameters are required for the optimisation process to begin. Reed parameters are

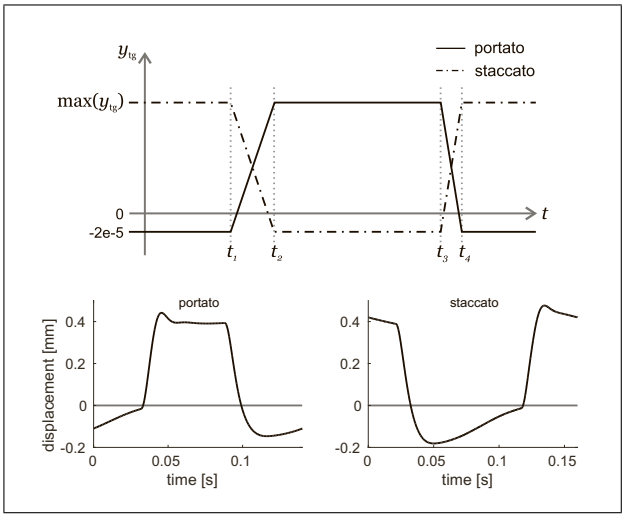


Figure 4. Top: Sketch of the temporal evolution of the tongue motion for modelling portato and staccato articulation. Similar curves are used as an input in LabView to instruct the motion of the tongue. Bottom: Measured tongue-shaker displacement when using the artificial blowing machine to generate portato and staccato.

taken from [23]. The initial value of the impact damping is set to $r_{\text{lay}} = 2$ s/m. The impact exponents are not optimised for and set to $\alpha_{\text{tg}} = 1.2$ and $\alpha_{\text{lay}} = 2$ [52]. Starting values for tongue-related parameters are obtained by visual inspection of the measured signals. The length of the bore is initially estimated by the clarinet geometry (using a cylinder-cone-cylinder model), the reed width is set to $\lambda = 1.2$ cm and the effective reed area to $S_{\text{r}} = 98.6\,\text{mm}^2$ [52]. Discretisation parameters are set to $\Delta_t = 1/44100$ s, while Δ_x is chosen such that—for the linear part of the model (resonator)—the Courant-Friedrichs-Lewy stability condition is satisfied [31] for the largest possible Courant number, with L being an integer multiple of Δ_x (here initially $\Delta_x = 7.96$ mm and changes accordingly when L is varied during optimisation). For the nonlinear excitation mechanism stability is ensured through conservation of $\mathcal{K}^n$ [45, 46].

Resynthesis results are given in Figure 6 for portato (case DP) and Figure 7 for staccato articulation (case Gp). The evolution of the objective function for both cases is shown in Figure 5 where the error between measured and resynthesised sound, defined as

$$\frac{\mathbf{p}-\hat{\mathbf{p}}}{||\mathbf{p}||}, \tag{44}$$

is also plotted. Using a different objective function could shift the emphasis towards higher frequencies, which might be more efficient in inverse modelling of brass instruments. The optimised physical model parameters are given in Table I. These parameters can be used to resynthesise the measured signals using the presented physical model. Note the effective nature of the reed parameters; even though the same reed is used across all experiments, its effective mass, damping and stiffness may vary since the properties of the reed-mouthpiece-embouchure

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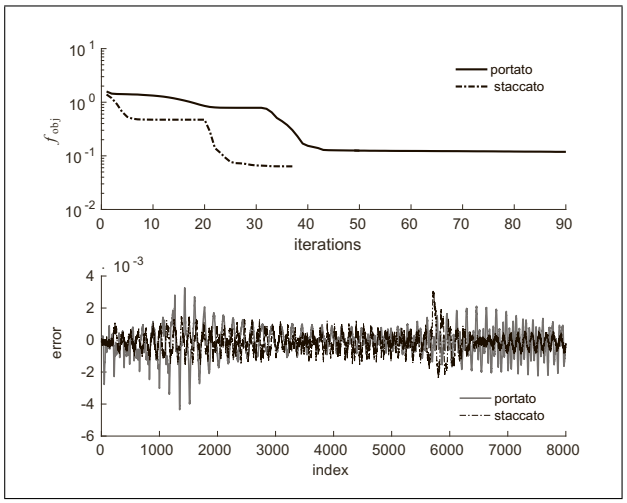


Figure 5. Top: Evolution of the objective function as defined in (43) during the optimisation process. Bottom: Difference between measured and resynthesised mouthpiece pressure as defined in (44).

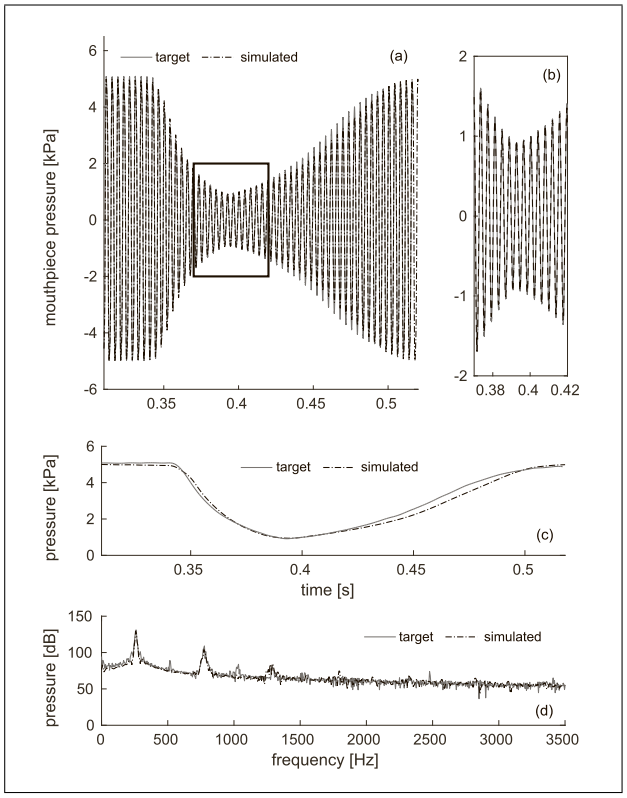


Figure 6. (a): Simulated mouthpiece pressure during portato articulation (case DP) compared to the measured signal obtained using the artificial blowing machine. (b): Zoom into the note transition inside the boxed region. (c): Envelope of mouthpiece pressure signals. (d): Spectra of mouthpiece pressure signals.

system, and not only the material properties of the reed, are represented by a single mass-spring oscillator.

It can be observed that the physical model is capable of reproducing both types of articulation carried out by the artificial player. The measured and simulated pressure signals present an almost perfect match in their time-domain

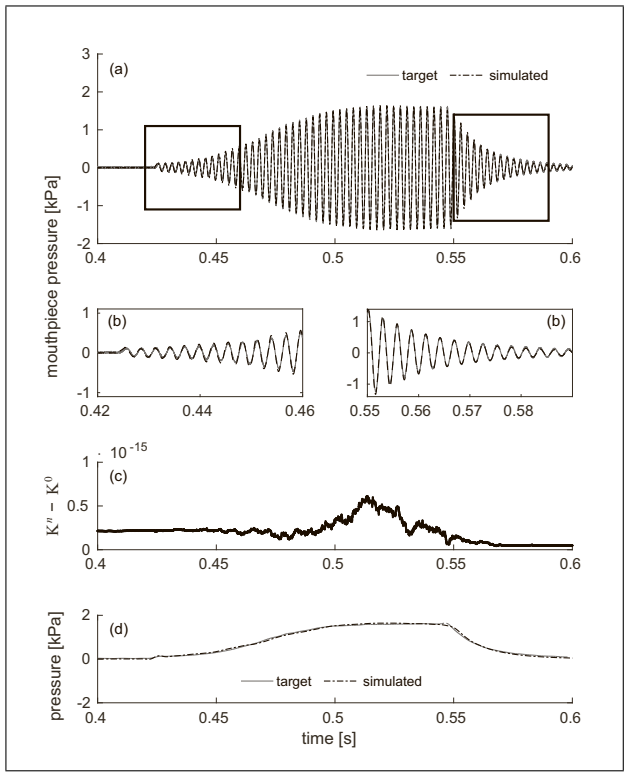


Figure 7. (a): Simulated mouthpiece pressure during staccato articulation (case Gp) compared to the measured signal obtained using the artificial blowing machine. (b): Zoom into the attack and release transients inside the boxed regions. (c): Error in the conservation of $\mathcal{K}$. (d): Envelope of mouthpiece pressure signals.

Table I. Initial and optimised parameters for portato (case DP) and staccato (case Gp) articulation, estimated using the signals obtained with the artificial blowing machine.
[a]Note that two different resonator lengths are used, hence two different starting values are given for the two notes played.
[b]from [6, 12, 23]
[c]Note that for staccato the tongue behaviour is inverted, i.e. tongue-reed contact occurs before t_1 and after t_4 and the reed is released in between. Hence two different starting values have been used for $t_1 \ldots t_4$ for portato / staccato articulation.

parameter	initial	portato	staccato	units
L	0.3294 / 0.2468[a]	0.3343	0.2466	m
m	3.25[b]	8.62	7.16	mg
k	1765[b]	1308	1713	N/m
γ	3000[b]	8190	10569	s^{-1}
y_{lay}	0.246[b]	0.385	0.256	mm
y_{eq}	0.41[b]	0.87	0.54	mm
k_{lay}	1.97×10^{6} [b]	6.16×10^{6}	8.53×10^{6}	N/m^2
r_{lay}	2	1.7	2.1	s/m
k_{tg}	3.16×10^{4}	1.06×10^{4}	1.14×10^{4}	$N/m^{\alpha_{\mathrm{tg}}}$
r_{tg}	2	1.5	2.1	s/m
$\max(y_{\mathrm{tg}})$	0.51	0.63	0.41	mm
t_1[c]	0.335 / 0.424	0.334	0.424	s
t_2[c]	0.34 / 0.429	0.353	0.430	s
t_3[c]	0.39 / 0.53	0.354	0.534	s
t_4[c]	0.45 / 0.56	0.54	0.559	s

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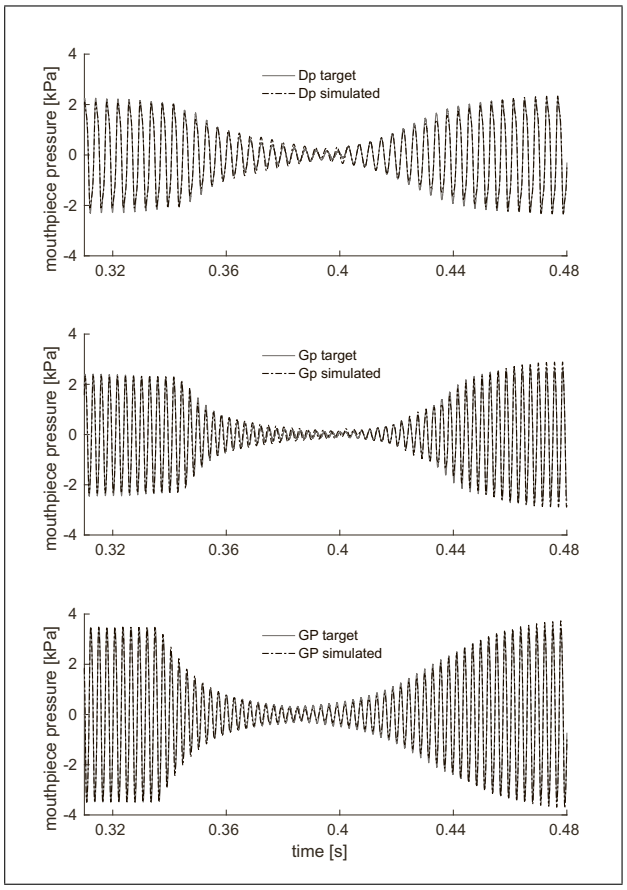


Figure 8. Measured and simulated mouthpiece pressure for the remaining combinations of fingering and blowing pressure for portato articulation.

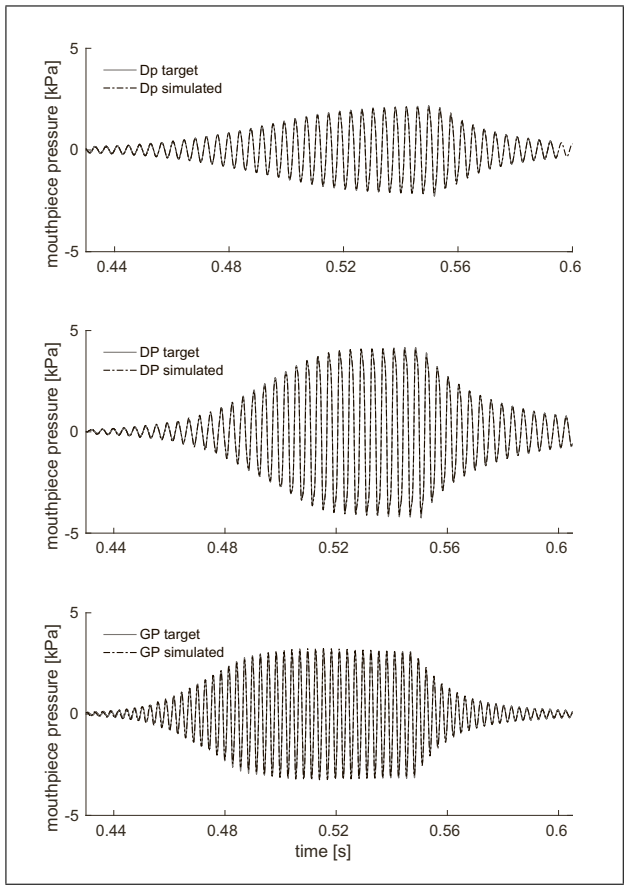


Figure 9. Measured and simulated mouthpiece pressure for the remaining combinations of fingering and blowing pressure for staccato articulation.

waveforms, which highlights the ability of the physical model to resemble the experimental setup. This implies that all relevant physical phenomena are taken into account during the formulation of the model and no significant effects are overlooked. Even though a lumped model is used to estimate the reed motion, the complexity added to the model by considering nonlinear interactions involving both the tongue and the mouthpiece is sufficient for simulating transients, while maintaining computational efficiency. Indeed, the model runs faster than real time on a standard desktop computer (Intel® Xeon(R) CPU E5-1650; 16Gb RAM). Furthermore, deviations in the conserved quantity $\mathcal{K}$ are in the order of machine precision, due to quantisation errors in finite-precision arithmetic (see Figure 7c). Hence no artificially generated energy is added to or subtracted from the system.

A small difference in the mouthpiece pressure envelope during the attack transient can be observed for the portato articulation (see Figure 6 at 0.45 s). A likely cause for that is the fact that no attempt has been made in this work to refine the model of the viscothermal losses, or optimise the relevant parameters. This subject is still under study, particularly in the case of wooden resonators [57]. It is also interesting to observe that, while odd harmonics are completely suppressed in the simulated signal, due to the cylindrical bore geometry, they weakly appear in the measurements. This could be attributed to the fact that the simulated bore geometry is not identical with the clarinet bore used in the measurements. The actual bore deviates from a strictly cylindrical geometry, due to closed toneholes, discontinuities at the joints and other irregularities during manufacture, whereas the virtual instrument is terminated at its first open tonehole, thus any effects of an open tonhole array are neglected. This is expected to be of minor significance in the lower register of the clarinet [18], where the measurements are performed.

Figures 8 and 9 show measured and resynthesised signals for all other combinations of articulation, fingering and blowing pressure. The values of the extracted reed parameters are given in Appendix A2. Sound examples for both types of articulation can be found on the accompanying webpage
https://resolver.obvsg.at/resolver?identifier=urn%3Anbn%3Aat%3Aat-ubmw-20180219201015213-1569042-2&verb=full.

6. Concluding Remarks

This article has explored time-domain simulations of single-reed woodwind instruments with special attention given to transient phenomena. To this cause a physical model of the nonlinear excitation mechanism was formulated, that takes into account the actions of the player's

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tongue. To ensure numerical stability an energy-based approach is adopted, leading to an unconditionally stable algorithm for the player-reed interaction.

In order to validate the model, measurements have been carried out using an artificial blowing machine. An artificial tongue capable of reproducing a variety of articulatory actions has been incorporated to the machine. Using an optimisation algorithm to estimate physical model parameters, signals quantitatively similar to the measured ones have been successfully synthesised. Since the measured signals closely resemble portato and staccato articulation performed by real players [14] the presented model may be used to resynthesise signals captured under real playing conditions. Future applications, including physics-based sound synthesis, performance analysis, and parameter estimation, could benefit from such a model.

The success of this inverse modelling procedure relies on the accurate measurements of both the reed vibrations and the internal mouthpiece pressure. Although these signals can be similarly obtained under real-playing conditions [23], this limits the player to a given range of mouthpiece-reed combinations. A more far-reaching, yet challenging approach would require only measurements of the radiated sound in order to estimate the model parameters. But this in turn calls for a sufficiently accurate model of sound radiation, as well as measurements carried out in an anechoic environment.

Acknowledgments

The authors acknowledge Alexander Mayer for his indispensable efforts on the construction of the artificial blowing machine. This research is supported by the Austrian Science Fund (FWF): P28655-N32.

Appendix

A1. Calculation of the energy balance

From (20b), it can be deduced that

$$Q_{\mathrm{in}} = \lambda h \sqrt{\frac{2}{\rho}} |p_\Delta|^{1/2} p_\Delta \operatorname{sign}(p_\Delta) = u_{\mathrm{f}} p_\Delta, \tag{A1}$$

hence Equation (24) can be written as

$$\frac{\mathrm{d}\mathcal{H}_{\mathrm{R}}}{\mathrm{d}t} = -Q_{\mathrm{r}} - u_{\mathrm{f}} p_\Delta - p_{\mathrm{in}} u_{\mathrm{in}} + p_{\mathrm{b}} u_{\mathrm{in}} + \frac{\varphi}{m} f_{\mathrm{tg}}. \tag{A2}$$

Now,

$$\begin{aligned} -u_{\mathrm{f}} p_\Delta + p_{\mathrm{b}} u_{\mathrm{in}} &= (p_{\mathrm{in}} - p_{\mathrm{b}})(u_{\mathrm{in}} - u_{\mathrm{r}}) + p_{\mathrm{b}} u_{\mathrm{in}} \\ &= p_{\mathrm{in}} u_{\mathrm{in}} - p_{\mathrm{in}} u_{\mathrm{r}} + p_{\mathrm{b}} u_{\mathrm{r}} = u_{\mathrm{r}} p_\Delta + p_{\mathrm{in}} u_{\mathrm{in}}, \end{aligned} \tag{A3}$$

hence

$$\begin{aligned} \frac{\mathrm{d}\mathcal{H}_{\mathrm{R}}}{\mathrm{d}t} &= -Q_{\mathrm{r}} + u_{\mathrm{r}} p_\Delta + \frac{\varphi}{m} f_{\mathrm{tg}} \\ &= -Q_{\mathrm{r}} + S_{\mathrm{r}} \frac{\mathrm{d}y}{\mathrm{d}t} p_\Delta + \frac{\varphi}{m} f_{\mathrm{tg}} \end{aligned} \tag{A4}$$

Table A1. Estimated parameters for the signals obtained with the artificial blowing machine. On the *case* column D and G stand for the note (261 Hz and 349 Hz respectively), p stands for low blowing pressure (4500 Pa) and P for high blowing pressure (6500 Pa).

parameter	portato	staccato	units	case
mass m	7.77	6.36	mg	Dp
	8.62	4.42	mg	DP
	8.43	7.16	mg	Gp
	4.30	3.54	mg	GP
stiffness k	1732	1491	N/m	Dp
	1308	1167	N/m	DP
	1757	1713	N/m	Gp
	1829	1564	N/m	GP
damping γ	8320	10271	s^{-1}	Dp
	8190	9084	s^{-1}	DP
	7197	10569	s^{-1}	Gp
	23424	30896	s^{-1}	GP

and

$$\begin{aligned} \frac{\mathrm{d}\mathcal{H}}{\mathrm{d}t} &= -Q_v - Q_\theta - Q_{\mathrm{r}} - Q_L \\ &\quad + \frac{\mathrm{d}y}{\mathrm{d}t} S_{\mathrm{r}} p_\Delta + \frac{\varphi}{m} f_{\mathrm{tg}} + p_{\mathrm{in}} u_{\mathrm{in}} \end{aligned} \tag{A5}$$

which, by substituting $\mathrm{d}y/\mathrm{d}t = \varphi/m$ and $S_{\mathrm{r}} p_\Delta = f_\Delta$ leads to Equation (26).

A2. Estimated parameters

The reed parameters estimated using the inverse model for eight different measurements (2 articulation techniques, 2 fingerings, 2 blowing pressures) are listed in Table A1.

It can be observed that the estimated effective mass and stiffness of the reed are lower for staccato compared to portato articulation, while the damping is higher. Furthermore for the case of the higher note played with a high blowing pressure (GP) the effective damping of the reed is approximately three times higher than in the other cases both for portato and for staccato articulation. These parameters are here estimated in the context of sound resynthesis. Since the measured signals required by the inverse model could also be measured for human players, a similar methodology could be carried out in an attempt to analyse differences across players and playing techniques.

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